

Uncertainty in AI

Uncertainty is an inevitable component in everyone's life. The morning weather report informs us that there is a 30% chance of evening showers. The business section in your newspaper reports that there is a 50% chance that the housing foreclosure crisis in your community will get worse before it improves. Naturally, if our AI systems are going to be robust, they must possess the capability to contend with such uncertainties.

Fuzzy logic and probability theory are two often-used tools. Fuzzy logic assigns grayness levels to events that were previously declared to be black or white. For example, when it is raining, the traction control system on your new car should engage. Suppose there is a light drizzle, then it is raining to a certain extent. Fuzzy logic supplies the wherewithal to contend with such uncertainties.

Fuzzy Sets and Fuzzy Logic

Suppose your instructor asked you to raise your hand if you are male then asked you to put your hands down. Next you were asked to raise your hand if you are female. Undoubtedly, each student in your class raised their hands once and only once. The following sets:

$M = \{x \mid x \text{ is a male student in your class}\}$

$F = \{y \mid y \text{ is a female student in your class}\}$

are examples of crisp sets in that each student in your class belongs to one and only one set. The intersection of these two sets is empty, i.e.

$M \cap F = \emptyset$ meaning that no element is a member of both sets.

Imagine next that everyone in your class has a job; your instructor now asks you to raise your hand if you are satisfied with your jobs. Then you are asked to raise your hand if you are dissatisfied with your job. It is likely that some hands went up both times. Some hands may have even been only somewhat raised in each case. Job satisfaction can be thought of as a fuzzy concept in that most people are not entirely happy or unhappy with their jobs.

Another example of a fuzzy set is the set of all tall people. A person whose height is 7'6" is definitely tall and if someone is 4'5" they are not. What should be said of individuals who are 5'10"? Well, you can say they are tall to a certain extent.

“Tallness” is seen to be a ‘fuzzy concept. A person who is 5’ tall (or shorter) is not a member of the set of tall people. A person who is 6’ tall might have a degree of membership of 0.65 in this set; we would write this as $\mu_t(6') = 0.65$, where $\mu_t(\)$ is the membership function for this set. You certainly would agree that $\mu_t(7' 6'') = 1.0$.

Let X be a classical universal set.

A real function $\mu_A : X \rightarrow [0, 1]$ is the membership function for the set A . The set of all pairs $(x, \mu_A(x))$ defines the fuzzy subset A of X .

A membership function completely specifies a fuzzy set. The set of all elements x that belong to X where $(x, \mu_A(x))$ belongs to A and $\mu_A(x) > 0$ is called the set of support of the fuzzy set A . For the set t of all tall people, the set of support consists of all people 5’ in height or taller. If A is a set with a finite set of support $\{a_1, a_2, \dots, a_m\}$ then this can be represented as

$$A = \mu_1 / a_1 + \mu_2 / a_2 + \dots + \mu_m / a_m,$$

where $\mu_i = \mu_A(a_i)$, $i = 1, \dots, m$. Note, the “/” and “+” symbols are being used as delimiters, no divisions or additions are being performed. For example, if $X = \{x_1, x_2, x_3\}$ where A and B are the two (crisp) subsets: $A = \{x_1, x_3\}$ and $B = \{x_2, x_3\}$, then these sets can be represented as

$$A = 1 / x_1 + 0 / x_2 + 1 / x_3$$

$$B = 0 / x_1 + 1 / x_2 + 1 / x_3.$$

The union of two sets A and B denoted $A \cup B$, is the set of all elements that lie in either A or B (or both). $A \cup B$ can be computed by taking for each x_i , the maximum of its membership in either set, i.e.,

$$A \cup B = 1 / x_1 + 1 / x_2 + 1 / x_3.$$

This methodology easily generalizes to the case in which the sets are fuzzy. For example, if

$$C = 0.2 / x_1 + 0.5 / x_2 + 0.8 / x_3$$

$$D = 0.6 / x_1 + 0.4 / x_2 + 0.2 / x_3,$$

Then the fuzzy union of C with D is

$$C \cup D = 0.6 / x_1 + 0.5 / x_2 + 0.8 / x_3.$$

The fuzzy intersection of two sets can be defined by using the minimum instead of the maximum of each degree of membership. So, for our previous example,

$$C \cap D = 0.2 / x_1 + 0.4 / x_2 + 0.2 / x_3$$

The complement of the crisp set E, i.e., E^c is the set of all elements that lie in the universal set (X in this example) but not in E. The complement E^c where E is a fuzzy set can be computed by

$$\mu_{E^c}(x) = 1 - \mu_E(x), \forall x \in X.$$

For example, if E is the fuzzy subset,

$$E = 0.3 / x_1 + 0.1 / x_2 + 0.9 / x_3,$$

then the complement of E equals

$$E^c = 0.7 / x_1 + 0.9 / x_2 + 0.1 / x_3.$$

Observe that, in general, when A is a fuzzy set, the union of A with its complement does not equal the universal set, and the intersection of A with its complement does not equal the empty set, as is the case for crisp sets. For the fuzzy set E we have

$$E \cup E^c = 0.7 / x_1 + 0.9 / x_2 + 0.9 / x_3$$

$$E \cap E^c = 0.3 / x_1 + 0.1 / x_2 + 0.1 / x_3$$

Fuzzy Inferences

Production systems and Expert Systems show how knowledge-based systems can be used to solve real-world problems, such as why your car won't start or what disease you might have contracted. Fuzzy measures can be applied to production rules to reflect the vagueness that is present in the world. For example, perhaps your car will not start because the battery is dead, or maybe the fuel tank is empty, or, if it's too cold, the motor could have frozen.

Fuzzy production rules have the same structure as their more traditional counterparts, for example,

Rule1: If $(A \bar{\vee} B)$, then C.

Rule2: If $(A \bar{\wedge} B)$, then D.

Suppose that the truth values of A and B are 0.1 and 0.8, respectively. Then

$$A \vee B = (0.1, 0.8) = 0.8$$

$$A \wedge B = \min (0.1, 0.8) = 0.1$$

Rules 1 and 2 can be applied only to some extent. Rule 1 is applied 80% and Rule 2 is applied 10%, in order that a combination of actions (or deductions) C and D will occur.

PROBABILITY THEORY AND UNCERTAINTY

Some would contend that probability theory witnessed its inception in 1654. A friend of Blaise Pascal was interested in gambling problems—what resulted was a series of mathematical communications between Pascal and Pierre de Fermat. It should come as no surprise then, that probability theory plays a prominent role in handling uncertainty.

Consider the experiment of tossing a fair coin twice. The sample space for an experiment S is the set of all possible outcomes (an outcome is sometimes referred to as a set of sample points). In our case, S, in which a coin is tossed twice, equals {(H, H), (T, T), (T, H), (H, T)}.

The probability of an event E_i is defined as:

$P(E_i)$ = the number of ways in which E_i can occur, divided by the total number of possible outcomes.

For example, the probability of occurrence of Tail followed by either Tail or head equals:

Here $E_3 = \{(T, T), (T, H)\}$

$P(E_3) = 2/4$ or $1/2$, as when a fair coin is tossed, two sample points correspond to this event, whereas |S| equals 4.

There are three basic axioms of probability measures:

- For any event E : $P(E) \geq 0$
- $P(S) = 1$ // when two coins are tossed, some outcome is certain to occur.
- If the events E_1 and E_2 are mutually exclusive, then $P(E_1 \cup E_2) = P(E_1) + P(E_2)$

// For example, if E1 equals two heads occurring when a coin is tossed twice, and E2 corresponds to two tails occurring, then $E1 \cup E2$ is the event corresponding to either two heads or two tails occurring. The probability of this event equals

$$P(E1 \cup E2) = P(E1) + P(E2) = 1/4 + 1/4 = 1/2.$$

A function that satisfies these three axioms is referred to as a probability function.

Bayes Theorem

In statistics and probability theory, the Bayes' theorem (also known as the Bayes' rule) is a mathematical formula used to determine the conditional probability of events. Essentially, the Bayes' theorem describes the probability of an event based on prior knowledge of the conditions that might be relevant to the event.

The theorem is named after English statistician, Thomas Bayes, who discovered the formula in 1763. It is considered the foundation of the special statistical inference approach called the Bayes' inference.

The Bayes' theorem is expressed in the following formula:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

Where:

$P(A|B)$ – the probability of event A occurring, given event B has occurred

$P(B|A)$ – the probability of event B occurring, given event A has occurred

$P(A)$ – the probability of event A

$P(B)$ – the probability of event B

Note that events A and B are independent events (i.e., the probability of the outcome of event A does not depend on the probability of the outcome of event B).

For example, You might be interested in finding out a patient's probability of having liver disease if they are an alcoholic. "Being an alcoholic" is the test (kind of like a litmus test) for liver disease.

A could mean the event “Patient has liver disease.” Past data tells you that 10% of patients entering your clinic have liver disease. $P(A) = 0.10$.

B could mean the litmus test that “Patient is an alcoholic.” Five percent of the clinic’s patients are alcoholics. $P(B) = 0.05$.

You might also know that among those patients diagnosed with liver disease, 7% are alcoholics. This is your $B|A$: the probability that a patient is alcoholic, given that they have liver disease, is 7%.

Bayes’ theorem tells you:

$$P(A|B) = (0.07 * 0.1)/0.05 = 0.14$$

In other words, if the patient is an alcoholic, their chances of having liver disease is 0.14 (14%). This is a large increase from the 10% suggested by past data. But it’s still unlikely that any particular patient has liver disease.